

The University of Zambia

Department of Economics

Introduction to Macroeconomic Theory (ECN 1215) Test 2

INSTRUCTIONS: ANSWER ALL QUESTIONS AND USE DIAGRAMS WHERE APPROPRIATE

TIME ALLOWED: 45 MINUTES

QUESTION 1

Given a closed economy with no government with the following consumption function

$$C = 100 + 0.75Y$$

Where Y represents National Income and C Consumption Expenditure

- a) Using this information, complete the table below (S denotes Savings) (5 marks)

Y	0	100	200	300	400	600	700	800	900	1000
C	100	175	250	325	400	550	625	700	775	850
I	50	50	50	50	50	50	50	50	50	50
S	-100	-75	-50	-25	0	50	75	100	125	250

- b) Interpret the above consumption function (5 marks)
 c) From part (a) above, compute the marginal propensity to save and interpret your answer (5 marks)
 d) Using the data from the above table, draw the savings function, consumption function and Aggregate Demand Schedule (10 Marks)
 e) From your diagram in part (c) above, what is the equilibrium level of income? (5 marks)

$$AD = C + I$$

$$175 + 50$$

$$C = 100 + 0.75Y$$

$$225$$

QUESTION 2

- a) Explain with the aid of a diagram, the significance of the 45° line in the Keynesian macroeconomic model. What do points above and below the 45° line represent? (5 marks)
 b) Given a closed economy without a government sector, calculate the equilibrium level of income of this economy if the level of autonomous consumption is K26 million, the slope of the curve is $\frac{4}{5}$ and the level of investment which is also autonomous is K10 million (5 marks)
 c) Use a diagram to explain how a decrease in investment spending will affect the equilibrium level of income in a Keynesian model of an economy without a government and a foreign sector. Comment on the relationship between the size of the change in the investment spending and the size of the change in the equilibrium level of income. (10 marks)

Total Marks: 50 marks

The End

Good Luck

$$S = -100 + 0.25Y$$

THE UNIVERSITY OF ZAMBIA
SCHOOL OF HUMANITIES AND SOCIAL SCIENCES
DEPARTMENT OF ECONOMICS

COURSE CODE : ECN 1215

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TUTOR : MR. CHITUMBO S. BUPE.

TUTORIAL GROUP : 15-16 hours Thursday

LECTURER : MR. BANDA. C.

DATE : 20th December 2013

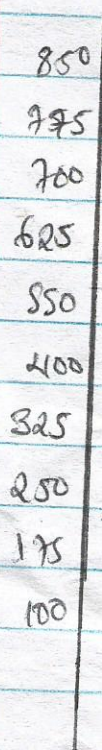
18.25
50

1 (a) $C = 100 + 0.75Y$

Y	0	100	200	300	400	500	600	700	800	900	1000
C	100	175	250	325	400	475	550	625	700	775	850
I	50	50	50	50	50	50	50	50	50	50	50
S	-100	-75	-50	-25	0	25	50	75	100	125	150

4.75

(b)



Cancelled

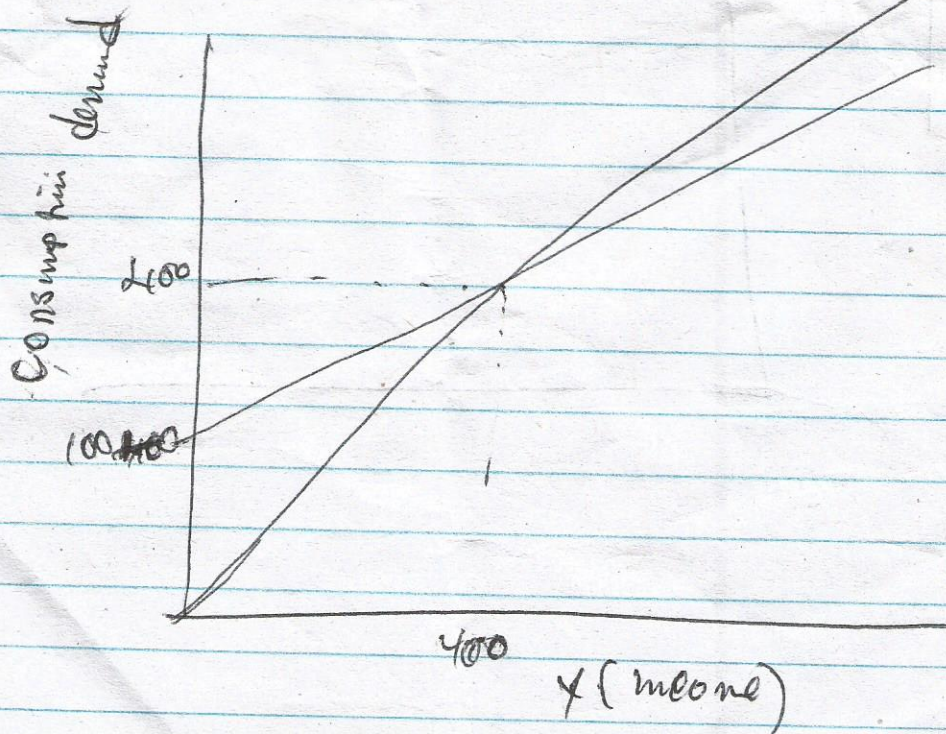
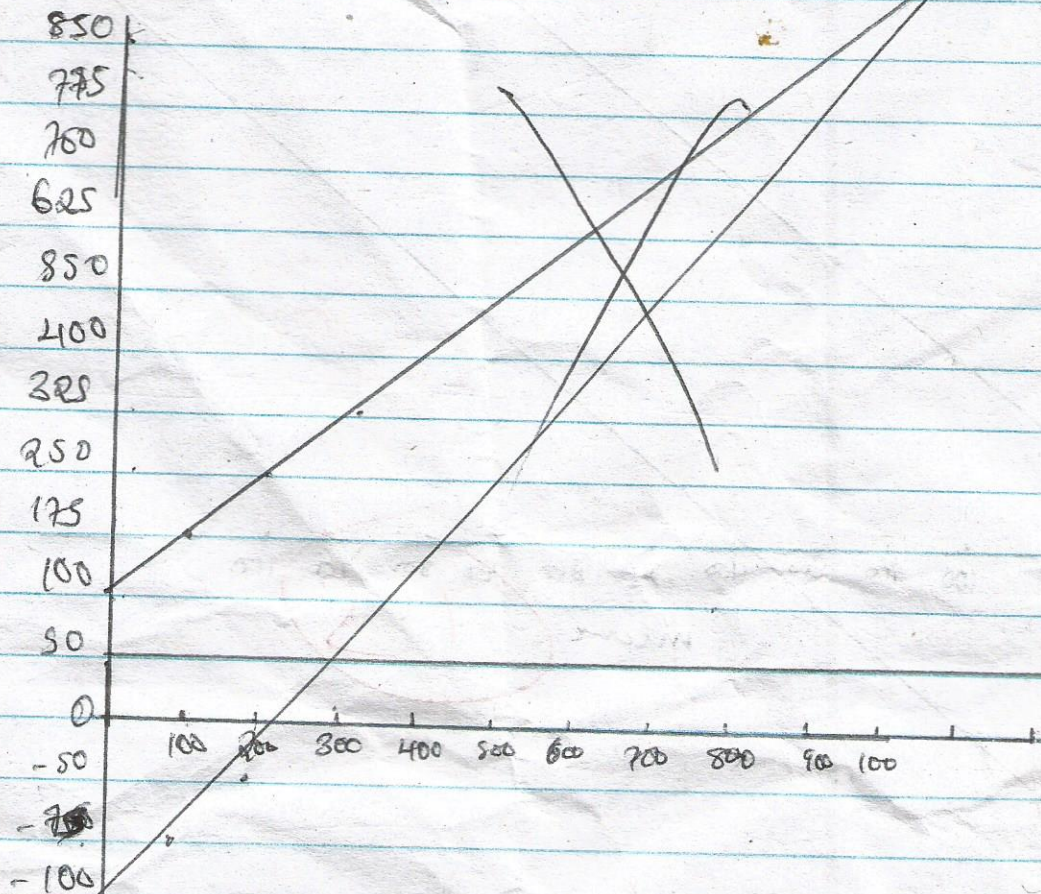
(b) From the above Consumption function ($C = 100 + 0.75Y$), it is indicated that with a Zero income the Autonomous Consumption Expenditure is 100. As Income (Y) increases, the Consumption also increase. Conversely, An increase in income also shows an increase in saving. Investment does not depend on the income as indicated in the table above.

(c) The Marginal propensity to save is the ratio of an earned income not intended to be consumed.

APSS = According to the Consumption function given ($C = 100 + 0.75Y$). When Autonomous Consumption is 100, Saving is 50.

and the marginal propensity to consume is 0.25 of the income earned.

$$MPS = 0.25$$



Consumption

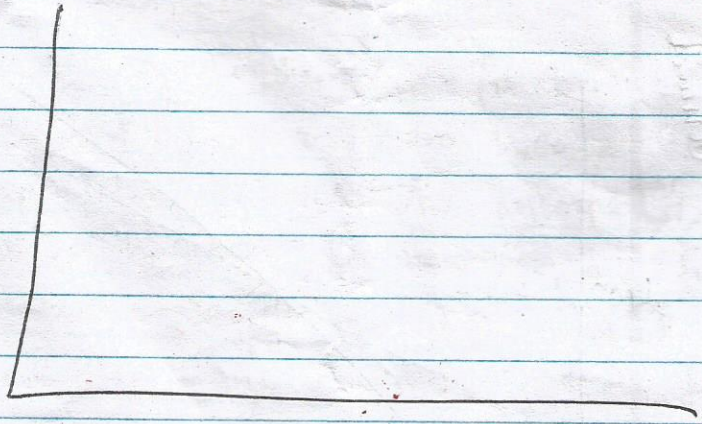
Savings

850
725
700
625
550
400
325
280
175
100
50
0
-25
-50
-75
-100

100 200 300 400 500 600 700 800 900 1000

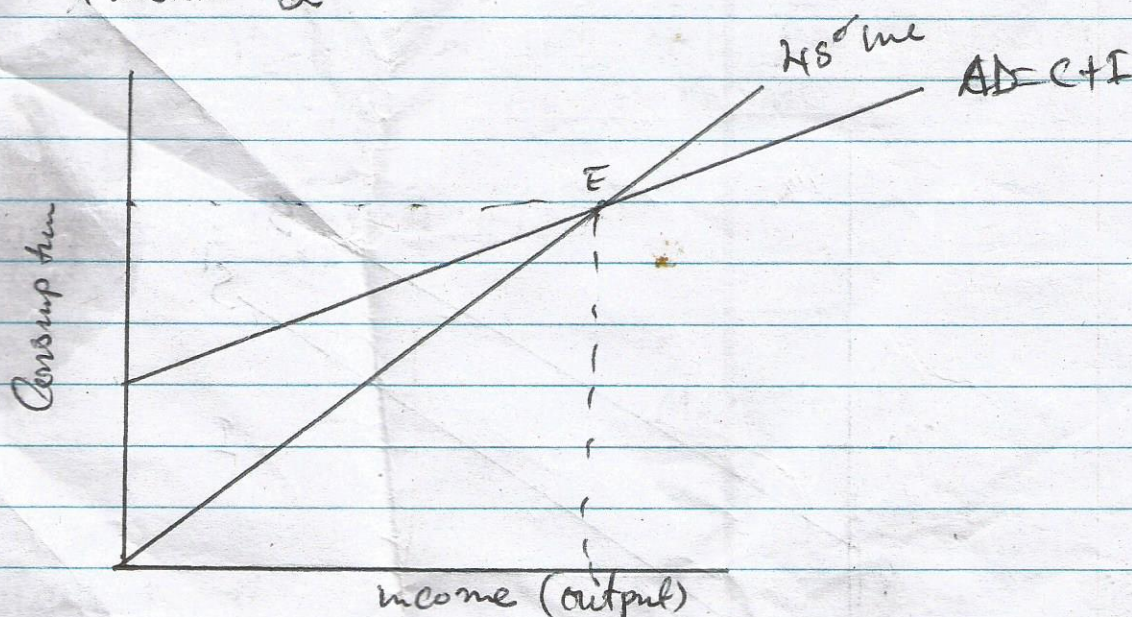
Income

1.5



Question 2

2)



The 45° line shows the levels of equal consumption to output or income. It represents ~~$C = Y$~~ on the diagram above.

The points above the 45° line show that demand aggregate demand is higher than the output. The points below the 45° line indicate high output and low demand. E is the level of equilibrium; output equals aggregate demand.

(b) $A = 26$ million

$MPC = 4/5$

$I = 10$ million

$$Y^* = \frac{A + I}{1 - C}$$

$$Y^* = \frac{26 + 10}{1 - 4/5}$$

S

$$= \frac{36}{1 - 0.8} = \frac{36}{0.2} = 180$$

$\therefore 180$ million is the equilibrium level of output

c.

